

Adaptive Daily Decision Engine

- Strict Behavior Coach System
- Full Project Process

Project Goal

- Help users make disciplined decisions
- Push users toward what they should do
- Adapt behavior over time

Problem Statement

- Users procrastinate and choose comfort options.
- Existing apps track tasks but do not enforce behavior.
- Need for a behavior-aware system.

Solution Overview

- A strict behavior coach system.
- Combines preferences, goals, and past behavior.
- Provides adaptive recommendations and feedback.

Tech Stack

- Java (TeaVM) for logic
- HTML/CSS for UI
- Minimal JavaScript
- LocalStorage for persistence
- No backend required

Core Components

- Decision Session
- Option & Criteria System
- Behavior Profile
- Goal System
- Scoring Engine

Data Model

- DecisionSession: type, options, criteria, chosenOption, timestamp
- Option: name, values (criterion->score)
- Criterion: name, weight
- BehaviorProfile: optionFrequency, timePatterns
- Goal: name, desiredWeights

Scoring Engine

- Base Score = weighted sum of normalized values
- Behavior Adjustment = frequency-based boost/penalty
- Goal Alignment = reward or penalty based on user goals
- Discipline Boost for hard options
- Comfort Penalty for easy options
- Final Score = base + behavior + goal + discipline/penalty

User Flow

- 1. Home Screen (status, streaks, failures)
- 2. Decision Template Selection
- 3. Input Options and Criteria
- 4. Result Screen (recommendation, explanation, confidence)
- 5. Feedback Confirmation
- 6. Insights & History Screens

UI Design Principles

- Minimal options to reduce friction
- Clear, direct language
- Strong color coding (Red=Failure, Green=Discipline, Yellow=Warning)
- Fast interaction (under 20 seconds per decision)
- Emotional triggers in messages

Enforcement System

- Streak tracking (disciplined choices)
- Failure logging (ignored recommendations)
- Feedback loop adjusts future scoring
- Brings accountability to each decision

Insights & Analytics

- Track comfort vs discipline choices
- Identify time-based patterns
- Show goal alignment percentages
- Provide brutal honesty feedback

Constraints

- Max 5 options
- Max 5 criteria
- Weights must sum to 1
- No missing values allowed

Build Plan / Phases

- Phase 1: Basic UI Skeleton
- Phase 2: Core Scoring Logic
- Phase 3: LocalStorage Persistence
- Phase 4: Behavior Tracking
- Phase 5: Adaptive Scoring
- Phase 6: Feedback & Insights
- Phase 7: UI Polish

Testing & Validation

- Test edge cases (equal scores, extreme weights)
- Conflict between goals and choices
- Missing data handling
- Behavior adaptation over time

Failure Risks

- Too soft → system ignored
- Too harsh → user quits
- Poor UX → abandonment
- Lack of feedback loop → no adaptation

Expected Outcome

- Users develop disciplined behavior
- Reduce decision fatigue
- Real-time feedback creates habit-forming engagement
- Insights help self-awareness

Demo / Scenario (Optional)

- Example decision: Study vs Relax
- Show recommendation, behavior penalty, discipline boost
- Feedback loop result

Contact / Credits

- Your Name / Team / References / Date